

Notes: Busse (RFS, 1999)
Volatility Timing in Mutual Funds: Evidence from Daily Returns

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1. Big picture

Busse studies whether actively managed equity mutual funds change market exposure when market volatility changes. The main question is not the traditional one of whether managers forecast market returns. It is whether they manage systematic risk in response to predictable variation in market risk.

The central result is that mutual funds reduce market exposure when conditional market volatility is high. This effect is especially strong among surviving funds and among growth-oriented funds. Busse also shows that this behavior is associated with higher contemporaneous risk-adjusted performance, especially higher Sharpe ratios and positive conditional alphas in high-volatility periods.

1.1. Incremental contribution

The paper contributes in four main ways.

1. It shifts the mutual fund timing question from *return timing* to *volatility timing*. Earlier work mostly asked whether funds raise beta before market upturns and lower beta before downturns. Busse asks whether funds lower beta when market volatility is high.
2. It is the first mutual fund study in this literature to use daily returns. Daily frequency improves estimation of changing factor exposures relative to monthly returns.
3. It controls for passive beta-volatility effects using random portfolios designed to match the funds' equity style exposures. This is important because stock betas can mechanically vary with market volatility even if managers do nothing.
4. It links volatility timing to fund performance. Funds that reduce market beta in high-volatility states have better contemporaneous Sharpe ratios, and fund alphas are higher when conditional volatility is high.

1.2. Main takeaway

The evidence supports the view that mutual funds condition their market risk exposure on market volatility. The paper does not prove that funds forecast volatility better than public conditional volatility models at monthly horizons. Instead, the strongest interpretation is that funds respond to volatility-related information in a way that improves risk-adjusted returns in high-volatility states.

2. Economic setting and intuition

Most market timing studies focus on expected returns. A manager is a successful return timer if the fund's market beta is higher before market upswings and lower before market downturns. Busse argues that this is not the only relevant timing problem. Since volatility is persistent and more forecastable than returns, managers may have a stronger opportunity to time *risk*.

The economic logic is straightforward. If expected market returns do not rise sufficiently when market volatility rises, then the optimal market exposure should fall in high-volatility states. This is precisely the comparative static Busse derives from a fund manager's expected utility maximization problem.

3. Model and derivations

3.1. Return-generating process

Assume a k -factor conditional return model for fund p :

$$R_{p,t+1} = \alpha_{pt} + \sum_{j=1}^k \beta_{jpt} R_{j,t+1} + \varepsilon_{p,t+1}, \quad (1)$$

where $R_{p,t+1}$ is the excess fund return, $R_{j,t+1}$ is factor j 's excess return, and β_{jpt} is chosen at time t .

The conditional expected return is

$$\mathbb{E}_t(R_{p,t+1}) = \alpha_{pt} + \sum_{j=1}^k \beta_{jpt} \mathbb{E}_t(R_{j,t+1}). \quad (2)$$

If the factors are orthogonal, the conditional variance is

$$\sigma_t^2(R_{p,t+1}) = \sum_{j=1}^k \beta_{jpt}^2 \sigma_{j,t+1}^2 + \sigma_t^2(\varepsilon_{p,t+1}). \quad (3)$$

3.2. Manager optimization

The manager chooses factor exposures to maximize expected utility:

$$\max_{\beta_{1pt}, \dots, \beta_{kpt}} \mathbb{E}_t [U_{t+1}(R_{p,t+1})]. \quad (4)$$

Differentiating with respect to β_{jpt} gives

$$0 = \frac{\partial}{\partial \beta_{jpt}} \mathbb{E}_t [U_{t+1}(R_{p,t+1})] \quad (5)$$

$$= \mathbb{E}_t [U'_{t+1}(R_{p,t+1}) R_{j,t+1}]. \quad (6)$$

Using the covariance decomposition,

$$\mathbb{E}_t [U' R_j] = \mathbb{E}_t [U'] \mathbb{E}_t [R_j] + \text{Cov}_t(U', R_j). \quad (7)$$

With conditional normality, Stein's lemma implies

$$\text{Cov}_t(U'(R_p), R_j) = \mathbb{E}_t [U''(R_p)] \text{Cov}_t(R_p, R_j). \quad (8)$$

Since $\text{Cov}_t(R_p, R_j) = \beta_{jpt} \sigma_{j,t+1}^2$ under orthogonal factors, the first-order condition becomes

$$\mathbb{E}_t [U'] \mathbb{E}_t [R_j] + \mathbb{E}_t [U''] \beta_{jpt} \sigma_{j,t+1}^2 = 0. \quad (9)$$

Define the Rubinstein risk-aversion measure

$$a = - \frac{\mathbb{E}_t [U''(R_{p,t+1})]}{\mathbb{E}_t [U'(R_{p,t+1})]}. \quad (10)$$

Then the optimal factor exposure is

$$\boxed{\beta_{jpt}^* = \frac{1}{a} \frac{\mathbb{E}_t(R_{j,t+1})}{\sigma_{j,t+1}^2}}. \quad (11)$$

3.3. When should beta decline with volatility?

Differentiate the optimal beta with respect to $\sigma_{j,t+1}$:

$$\frac{\partial \beta_{jpt}^*}{\partial \sigma_{j,t+1}} = \frac{1}{a\sigma_{j,t+1}^2} \left[\frac{\partial \mathbb{E}_t(R_{j,t+1})}{\partial \sigma_{j,t+1}} - \frac{2\mathbb{E}_t(R_{j,t+1})}{\sigma_{j,t+1}} \right]. \quad (12)$$

Therefore, a fund should reduce exposure to factor j when factor volatility rises if

$$\frac{\partial \mathbb{E}_t(R_{j,t+1})}{\partial \sigma_{j,t+1}} < \frac{2\mathbb{E}_t(R_{j,t+1})}{\sigma_{j,t+1}}. \quad (13)$$

A sufficient condition is

$$\frac{\partial \mathbb{E}_t(R_{j,t+1})}{\partial \sigma_{j,t+1}} \leq 0. \quad (14)$$

The paper emphasizes that empirical work does not find a reliably positive simple relation between expected market returns and conditional market volatility. Thus, a negative beta-volatility relation is economically plausible.

3.4. Passive beta-volatility effects

Busse also notes that even without active timing, an individual stock's market beta can mechanically vary with market volatility:

$$\beta_{mit} = \rho_{mit} \frac{\sigma_{it}}{\sigma_{mt}}. \quad (15)$$

This mechanical relation is why the paper compares mutual funds to random portfolios that mimic fund styles. The main inference is based on whether funds' volatility coefficients are more negative than those of passive random portfolios.

4. Data and empirical design

4.1. Mutual fund sample

The sample contains 230 domestic equity mutual funds from January 2, 1985 through December 29, 1995. The funds are selected from Wiesenberger's *Mutual Funds Panorama* and include maximum capital gains, growth, and growth-and-income funds. Sector funds, balanced funds, and index funds are excluded.

Daily returns are constructed from daily net asset values and dividends:

$$r_{pt} = \frac{NAV_{pt} + D_{pt}}{NAV_{p,t-1}} - 1. \quad (16)$$

The paper includes nonsurviving funds where possible to reduce survivorship bias. This is important because nonsurviving funds differ meaningfully from surviving funds in performance and in volatility timing behavior.

4.2. Random control samples

The paper constructs two random control samples to distinguish active timing from passive effects. For each mutual fund, Busse estimates style exposures by solving a quadratic programming problem:

$$\min_{b_{p1}, \dots, b_{pk}} \text{Var} \left(r_{pt} - \sum_{i=1}^k b_{pi} r_{it} \right), \quad (17)$$

where r_{it} are returns on asset-class portfolios. The asset classes include size/book-to-market intersections and momentum/contrarian categories.

The first random sample mimics an equal-weighted equity strategy rebalanced annually. The second mimics a buy-and-hold strategy without rebalancing. The notes below focus on the annually rebalanced random sample, which is the main benchmark in the tables.

4.3. Factor models

The basic daily CAPM-style model includes a lagged market return to control for nonsynchronous trading:

$$R_{pt} = \alpha_p + \beta_{mp} R_{mt} + \beta_{lmp} R_{m,t-1} + \varepsilon_{pt}. \quad (18)$$

The three-index model adds Fama-French SMB and HML factors. The four-index model adds a momentum-minus-contrarian factor, MMC, analogous to the momentum factor used in Carhart-style performance models.

4.4. Daily volatility timing specification

The central volatility timing idea is that market beta is allowed to vary with market volatility:

$$\beta_{mpt} = \beta_{0mp} + \gamma_{mp}(\sigma_{mt} - \bar{\sigma}_m). \quad (19)$$

Substituting into the single-index model gives

$$R_{pt} = \alpha_p + \beta_{0mp} R_{mt} + \gamma_{mp}(\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \beta_{lmp} R_{m,t-1} + \varepsilon_{pt}. \quad (20)$$

A negative γ_{mp} means the fund lowers its market beta when conditional market volatility is above average. The multi-index version is

$$R_{pt} = \alpha_p + \sum_{j=1}^k [\beta_{0jp} R_{jt} + \beta_{ljp} R_{j,t-1}] + \gamma_{mp}(\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \varepsilon_{pt}. \quad (21)$$

The paper models daily conditional market volatility using an EGARCH(1,1) model with leverage:

$$\ln \sigma_{mt}^2 = a + a_1 \frac{|\varepsilon_{m,t-1}| + \gamma_1 \varepsilon_{m,t-1}}{\sigma_{m,t-1}} + b_1 \ln \sigma_{m,t-1}^2. \quad (22)$$

4.5. Monthly realized-beta specification

For monthly tests, Busse estimates a fund's market beta inside each month using daily returns and then regresses the monthly beta on realized market volatility:

$$\beta_{mpt} = \beta_{0mp} + \gamma_{mp}(\sigma_{mt} - \bar{\sigma}_m) + \varepsilon_{mpt}. \quad (23)$$

This specification directly asks whether estimated systematic risk is lower in high-volatility months.

5. Main empirical findings

5.1. Market factor dominates fund variance

The S&P 500 market factor explains the overwhelming majority of the variance captured by the four-factor model. The average contribution is 90.6%, and the median contribution is 95.5%. SMB, HML, and MMC are much smaller. This justifies focusing the timing tests on market volatility rather than on size, value, or momentum volatility.

5.2. Market return and volatility are not strongly positively related

Using monthly data, the correlation between S&P 500 returns and realized volatility is negative in the sample period. Decomposing realized volatility into a conditional component and a volatility shock shows that the negative monthly relation is driven by the shock component. Using daily EGARCH conditional volatility, the correlation between S&P 500 returns and conditional volatility is approximately zero. The CBOE implied volatility series is highly correlated with the EGARCH estimates, which supports the volatility proxy.

5.3. Funds lower market exposure in high-volatility states

The daily and monthly tests both show negative volatility timing coefficients relative to random control portfolios. The strongest evidence appears in the four-index model. Growth funds, maximum capital gains funds, and surviving funds show especially strong negative beta-volatility relations. Growth-and-income funds and nonsurviving funds show weaker evidence.

5.4. No strong evidence of superior monthly volatility forecasting

When Busse asks whether funds respond to volatility shocks not predicted by ARMA or public instruments, the evidence is mixed. This weakens the claim that funds have superior volatility forecasts beyond public information at monthly horizons. The stronger interpretation is that funds condition market exposure on volatility-related information, not necessarily that they have private volatility forecasts.

5.5. Volatility timing and performance

Volatility timers have better contemporaneous Sharpe ratios. The evidence does not show that volatility timing estimated over one six-month period predicts the next six months' performance.

However, conditional alpha tests show that fund alpha is significantly higher in high-volatility states, suggesting that active funds may provide a useful volatility hedge.

5.6. Robustness

The results survive several checks: instrumental variables for errors in realized volatility, controlling for Treynor-Mazuy market return timing, controlling for market timing based on public instruments, checking fund cash flows, and excluding the 1987 crash period. Excluding the 1987 crash period makes the main volatility timing results stronger rather than weaker.

6. Interpretation and implications

Busse's results are important because they reveal a form of active management not captured by traditional return-timing tests. A manager who cannot forecast market direction may still improve risk-adjusted performance by lowering exposure when volatility is high and expected returns do not compensate sufficiently for the added risk.

The paper also clarifies why a conditional framework matters. Static alphas and betas can miss important time variation in risk exposure. A fund may look ordinary under unconditional analysis while providing state-contingent risk reduction in bad volatility states.

The survivor evidence is also important. Surviving funds are much more sensitive to volatility than nonsurvivors. This suggests that volatility timing is related to fund quality, fund survival, or both. The paper does not fully disentangle whether successful volatility timing causes survival or whether surviving funds simply have other characteristics correlated with timing.

7. Figures and tables

7.1. Table 1. Summary statistics of mutual fund sample and random control samples; Table 2. Daily volatility timing results

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Table 1
Summary statistics of mutual fund sample and random control samples

Panel A: Return statistics														
Fund group	Number of funds	Return	Standard deviation	Skewness	Kurtosis									
All	230	0.0562%	0.898%	-2.51	48.68									
Maximum capital gains	68	0.053%	0.97%	-2.32	42.85									
Growth	107	0.0570	0.884	-2.34	44.05									
Growth and income	55	0.0557	0.865	-3.05	64.91									
Surviving	196	0.0574	0.879	-2.50	49.82									
Nonsurviving	34	0.0492	1.005	-2.51	42.16									
Random samples	230	0.0577	0.870	-2.61	51.00									
Rebalanced	230	0.0567	0.810	-2.41	47.90									
Panel B: Model parameters														
Fund group	Single-index				Three-index				Four-index					
	α	β_m	β_{SMB}	R^2	α	β_m	β_{SMB}	β_{HML}	R^2	α	β_m	β_{SMB}	β_{HML}	R^2
All	-0.0064%	0.921	0.770	-0.0009	0.826	0.436	-0.186	0.836	-0.0054%	0.826	0.445	-0.180	0.052	0.839
Max capital gains	-0.0088	0.981	0.694	-0.0010	0.838	0.655	-0.139	0.800	-0.0057	0.839	0.668	-0.129	0.084	0.805
Growth	-0.0066	0.921	0.781	-0.0045	0.831	0.419	-0.198	0.840	-0.0049	0.831	0.428	-0.192	0.048	0.843
Growth & income	-0.0032	0.845	0.844	-0.0060	0.800	0.197	0.028	0.871	-0.0061	0.800	0.201	0.030	0.020	0.874
Surviving	-0.0035	0.917	0.777	-0.0010	0.829	0.404	-0.170	0.837	-0.0035	0.830	0.413	-0.164	0.053	0.841
Nonsurviving	-0.0240	0.942	0.734	-0.0166	0.805	0.622	-0.280	0.825	-0.0167	0.805	0.630	-0.274	0.043	0.829
Random samples	-0.0030	0.880	0.815	-0.0062	0.771	0.488	-0.036	0.887	-0.0063	0.771	0.489	-0.035	0.007	0.888
Rebalanced	-0.0044	0.886	0.844	-0.0062	0.800	0.373	-0.045	0.891	-0.0061	0.799	0.371	-0.047	-0.007	0.891
Not Rebalanced	-0.0044	0.886	0.844	-0.0062	0.800	0.373	-0.045	0.891	-0.0061	0.799	0.371	-0.047	-0.007	0.891

The table shows the average daily return statistics of the mutual fund sample and the two random control samples. The first random control sample mimics an equal-weighted equity rebalancing strategy, and the second random control sample mimics an equal-weighted strategy without rebalancing. The betas are sums of the corresponding betas of the three and four index specifications. The table period is from January 1, 1985 to December 29, 1995.

$$R_{it} = \alpha_i + \beta_{im} R_{mt} + \beta_{iSMB} R_{SMB,t} + \beta_{iHML} R_{HML,t} + \varepsilon_{it} \quad (11)$$

The table shows the average daily summary statistics of the mutual fund sample and the two random control samples. The first random control sample mimics an equal-weighted strategy that rebalances annually, and the second random control sample mimics an equal-weighted strategy without rebalancing. The betas are sums of the contemporaneous and lag betas from the single-index model.

$$R_{pt} = \alpha_p + \beta_{mp0} R_{mt} + \beta_{mp1} R_{m,t-1} + \epsilon_{pt} \quad (11)$$

and three- and four-index specifications. The sample period is from January 2, 1985 to December 29, 1995.

Table 1. Summary statistics of mutual fund sample and random control samples

Table 2
Daily volatility timing results

Fund group	Single-index		Three-index		Four-index	
	γ_{mp}	t -statistic	γ_{mp}	t -statistic	γ_{mp}	t -statistic
Panel A: Sample coefficients						
All	0.83	1.63	-3.89	-7.12**	-5.39	-9.15**
Max capital gains	1.76	1.86	-5.41	-5.25**	-7.63	-6.60**
Growth	-0.50	-0.65	-4.97	-6.32**	-6.49	-7.98**
Growth and income	2.27	2.44	0.06	0.07	-0.53	-0.53
Survivors	0.33	0.66	-3.95	-7.26**	-5.48	-9.05**
Nonsurvivors	3.85	2.00	-3.52	-1.75	-4.89	-2.48*
Random sample	4.06	24.40	-0.66	-3.62**	-0.88	-4.99**
Panel B: Difference between mutual fund and random sample coefficients						
All	-3.23	-6.00**	-3.24	-5.62**	-4.51	-7.33**
Max capital gains	-2.30	-2.41*	-4.75	-4.54**	-6.74	-5.77**
Growth	-4.57	-5.75**	-4.32	-5.34**	-5.60	-6.73**
Growth and income	-1.80	-1.91	0.72	0.74	0.35	0.35
Survivors	-3.74	-7.14**	-3.30	-5.75**	-4.59	-7.28**
Nonsurvivors	-0.22	-0.11	-2.86	-1.42	-4.01	-2.02*

The table shows the average daily volatility coefficients for the single-index volatility timing model,

$$R_{pt} = \alpha_p + \beta_{mp0} R_{mt} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \beta_{lmp} R_{m,t-1} + \epsilon_{pt}, \quad (13)$$

and three- and four-index volatility timing models,

$$R_{pt} = \alpha_p + \sum_{j=1}^k [\beta_{0jp} R_{jt} + \beta_{1jp} R_{j,t-1}] + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \epsilon_{pt}. \quad (20)$$

I generate conditional volatility estimates from an EGARCH(1,1) with leverage specification of S&P 500 returns:

$$\begin{aligned} \epsilon_{mt} \mid \epsilon_{m,t-1}, \epsilon_{m,t-2}, \dots \sim t(0, \sigma_m^2) \\ \ln \sigma_m^2 = a + a_1 \frac{|\epsilon_{m,t-1}| + \gamma_1 \epsilon_{m,t-1}}{\sigma_{m,t-1}} + b_1 \ln \sigma_{m,t-1}^2. \end{aligned} \quad (19)$$

Panel A shows the sample coefficients, and Panel B shows the difference between the mutual fund coefficients and the coefficients of the random control sample. I base the significance of the sample coefficient t -statistics on a two-tailed test. I base the significance of the differences between the mutual fund and random sample coefficients on a difference-in-means test in the hypothesized direction. *, ** Significant at the 5% and 1% level, respectively. The mutual fund sample consists of 230 funds, and the random control sample consists of 230 equal-weighted, annually rebalanced portfolios. The sample period is from January 2, 1985 to December 29, 1995.

Table 2. Daily volatility timing results

Interpretation of Table 1. The sample contains 230 funds: 68 maximum capital gains funds, 107 growth funds, and 55 growth-and-income funds. The average daily return is about 0.056%, with substantial negative skewness and high kurtosis. Nonsurviving funds have lower average returns and worse alphas than surviving funds. The four-index coefficients show that funds have meaningful exposure to the market and SMB and tilt away from value stocks through negative HML coefficients. Random control portfolios are broadly similar but not identical, which motivates comparing fund coefficients to random portfolio coefficients rather than interpreting raw fund coefficients alone.

Interpretation of Table 2. Table 2 provides the daily volatility timing evidence. The key panel is Panel B, which subtracts random control sample coefficients from mutual fund coefficients. Negative differences mean funds reduce market exposure more than passive portfolios when conditional market volatility is high. For all funds, the four-index difference is -4.51 with a large negative t -statistic. Growth funds and surviving funds show especially strong negative differences. Growth-and-income funds do not show robust negative timing, consistent with their more conservative mandate.

7.2. Table 3. Monthly volatility timing results; Table 4. Conditional monthly volatility timing results

Table 3
Monthly volatility timing results

Fund group	Single-index		Three-index		Four-index	
	γ_{mp}	t -statistic	γ_{mp}	t -statistic	γ_{mp}	t -statistic
Panel A: Sample coefficients						
All	-3.04	-6.58**	-4.88	-11.44**	-5.92	-11.64**
Max capital gains	-4.35	-4.25**	-6.94	-7.62**	-7.45	-6.35**
Growth	-3.53	-5.57**	-5.24	-8.79**	-6.79	-10.16**
Growth and income	-0.47	-0.69	-1.63	-2.99**	-2.38	-3.52**
Survivors	-3.77	-8.63**	-4.98	-11.87**	-5.96	-11.33**
Nonsurvivors	1.34	0.78	-4.29	-2.65**	-5.72	-3.41**
Random sample	0.75	4.07	-2.37	-20.25**	-3.51	-18.08**
Panel B: Difference between mutual fund and random sample coefficients						
All	-3.78	-7.62**	-2.51	-5.67**	-2.42	-4.44**
Max capital gains	-5.10	-4.90**	-4.57	-4.98**	-3.94	-3.31**
Growth	-4.27	-6.48**	-2.87	-4.72**	-3.28	-4.72**
Growth and income	-1.21	-1.72	0.74	1.34	1.13	1.61
Survivors	-4.52	-9.53**	-2.61	-5.99**	-2.45	-4.38**
Nonsurvivors	0.59	0.34	-1.92	-1.18	-2.21	-1.31

The table shows the average monthly volatility coefficients for the single-, three-, and four-index models:

$$\beta_{mpt} = \beta_{0mp} + \gamma_{mp}(\sigma_{mt} - \bar{\sigma}_m) + \varepsilon_{mpt}. \quad (21)$$

I compute total monthly realized market volatility estimates using

$$\sigma_{mt} = \left[\sum_{i=1}^{n_t} (R_{mti} - \bar{R}_{mt})^2 \right]^{\frac{1}{2}}. \quad (17)$$

Panel A shows the sample coefficients and Panel B shows the difference between the mutual fund coefficients and the coefficients of the random control sample. I base the significance of the sample coefficient t -statistics on a two-tailed test. I base the significance of the differences between the mutual fund and random sample coefficients on a difference-in-means test in the hypothesized direction. *, ** Significant at the 5% and 1% level, respectively. The mutual fund sample consists of 230 funds, and the random control sample consists of 230 equal-weighted, annually rebalanced portfolios. The sample period is from January 2, 1985 to

Table 3. Monthly volatility timing results

Table 4
Conditional monthly volatility timing results

	ARMA			Instruments		
	Single-index	Three-index	Four-index	Single-index	Three-index	Four-index
Mutual fund coefficients						
	0.132 (8.94)	-0.050 (-4.25)**	-0.091 (-5.69)**	0.038 (2.05)	-0.144 (-9.77)**	-0.163 (-9.89)**
Random sample coefficients						
	0.333 (22.87)	-0.057 (-6.40)**	-0.137 (-11.48)**	0.234 (17.99)	-0.139 (-12.36)**	-0.190 (-14.81)**
Difference between mutual fund and random sample coefficients						
	-0.201 (-9.70)**	0.007 (0.47)	0.047 (2.34)	-0.196 (-8.68)**	-0.006 (-0.31)	0.028 (1.33)

The table shows the average γ_{mp} coefficients of regressions of the residual component of monthly realized market volatility against the lagged single-, three-, and four-index S&P 500 beta,

$$\varepsilon_{mt}^e = \sigma_{m0} + \gamma_{mp}\beta_{mp,t-1} + \varepsilon_{mt}. \quad (23)$$

I estimate the residual components of volatility from an ARMA(1,1) specification,

$$\sigma_{mt} = \phi\sigma_{m,t-1} + \theta\varepsilon_{m,t-1} + \varepsilon_{mt}, \quad (18)$$

and an instrumental variables specification,

$$\sigma_{mt} = \sigma_{m0} + \sum_{j=1}^5 \phi_j z_{j,t-1} + \varepsilon_{mt}. \quad (22)$$

I compute the monthly realized market volatility estimates from

$$\sigma_{mt} = \left[\sum_{i=1}^{n_t} (R_{mti} - \bar{R}_{mt})^2 \right]^{\frac{1}{2}}. \quad (17)$$

The instrumental variables model uses five conditioning variables: T-bill, dividend yield, term structure, default premium, and a January dummy. The table shows the mutual fund coefficients, the random sample coefficients, and the difference between the mutual fund and random sample coefficients. t -statistics are in parenthesis. I base the significance of the mutual fund and random sample coefficient t -statistics on a two-tailed test. I base the significance of the differences between the mutual fund and random sample coefficients on a difference-in-means test in the hypothesized direction. *, ** Significant at the 5% and 1% level, respectively. The mutual fund sample consists of 230 funds, and the random control sample consists of 230 equal-weighted, annually rebalanced portfolios. The sample period is from January 2, 1985 to December 29, 1995.

Table 4. Conditional monthly volatility timing results

Interpretation of Table 3. The monthly beta regressions confirm the daily results. Mutual funds' market betas decline when realized monthly market volatility is high, and the decline is stronger than in the random control sample. For all funds, the four-index difference is -2.42 and statistically significant. Survivors show a strong negative difference, while nonsurvivors do not differ significantly from random portfolios. This survivor contrast is one of the strongest patterns in the paper.

Interpretation of Table 4. Table 4 studies whether funds react to volatility shocks not captured by public conditional volatility models. The results are mixed. The single-index difference is significantly negative for both the ARMA and instrument-based specifications, but the three- and four-index differences are not reliably negative. This weakens a private-information interpretation. The finding is better read as evidence that funds manage exposure with respect to volatility, while the paper cannot show strong monthly evidence that they forecast volatility better than public variables.

7.3. Table 5. Relation between volatility timing and performance; Table 6. Conditional alphas

Table 5
Relation between volatility timing and performance

Panel A: Contemporaneous performance

	High Sharpe ratio	Low Sharpe ratio	
Volatility timer	620	552	1172
Not volatility timer	551	621	1172
	1171	1173	2344
Chi-square			7.89 ($p = .0587$)

Panel B: Predicted performance

B1: Sharpe ratio

	High Sharpe ratio	Low Sharpe ratio	
Volatility timer	1092	1155	2247
Not volatility timer	1153	1093	2246
	2245	2248	4493
Chi-square			3.26 ($p = .071$)

B2: Conditional alpha

	High conditional alpha	Low conditional alpha	
Volatility timer	1113	1133	2246
Not volatility timer	1132	1115	2247
	2245	2248	4493
Chi-square			0.27 ($p = .5320$)

The table shows the number of funds allotted to each of four cells in two-by-two contingency tables based on the volatility coefficient, γ_{mp} , in the four-index version of

$$R_{pt} = \alpha_p + \sum_{j=1}^k [\beta_{0jp} R_{jt} + \beta_{1jp} R_{j,t-1}] + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \varepsilon_{pt} \quad (20)$$

and the Sharpe ratio (panel A and panel B1),

$$S_{pt} = \frac{r_{pt} - r_{ft}}{\sigma_{pt}}, \quad (25)$$

or the conditional alpha α_{pt} (panel B2) in Equation (20). In Panel A, the volatility coefficients and Sharpe ratios are annual and contemporaneous. In Panel B, I compute the volatility coefficients over one six-month time period and the Sharpe ratios or conditional alphas over the next six-month time period. I generate the conditional volatility estimates from an EGARCH(1,1) with leverage specification of S&P 500 returns. I determine the p-values empirically by comparing the actual cell allotments to the distribution of 10,000 bootstrapped allotments constructed under the null hypothesis that no relation exists between volatility timing and performance. The sample consists of 230 mutual funds. The sample period is from January 1, 1985 to December 29, 1995.

Table 6
Conditional alphas

Single-index	Three-index	Four-index
Mutual fund coefficients		
0.315 (5.11)**	0.187 (3.18)**	0.289 (4.80)**
Random sample coefficients		
0.067 (3.17)**	0.027 (1.84)	0.041 (2.95)**
Difference between mutual fund and random coefficients		
0.249 (3.81)**	0.160 (2.65)**	0.248 (4.01)**

The table shows the average daily conditional alpha coefficients, ϕ_{mp} , for the four-index conditional alpha model,

$$R_{pt} = \alpha_{0p} + \phi_{mp} (\sigma_{mt} - \bar{\sigma}_m) + \sum_{j=1}^k (\beta_{0jp} R_{jt} + \beta_{1jp} R_{j,t-1}) + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \varepsilon_{pt}. \quad (26)$$

I generate the conditional volatility estimates from an EGARCH(1,1) with leverage specification of S&P 500 returns:

$$\varepsilon_{mt} | \varepsilon_{m,t-1}, \varepsilon_{m,t-2}, \dots \sim t(0, \sigma_{mt}^2) \\ \ln \sigma_{mt}^2 = a + a_1 \frac{|\varepsilon_{m,t-1}| + \gamma_1 \varepsilon_{m,t-1}}{\sigma_{m,t-1}} + b_1 \ln \sigma_{m,t-1}^2 \quad (19)$$

The table shows the mutual fund coefficients, the random sample coefficients, and the difference between the mutual fund and random sample coefficients. t -statistics are in parenthesis. I base the significance of the mutual fund and random sample coefficient t -statistics on a two-tailed test. I base the significance of the differences between the mutual fund and random sample coefficients on a difference-in-means test. *, ** Significant at the 5% and 1% level, respectively. The mutual fund sample consists of 230 funds, and the random control sample consists of 230 equal-weighted, annually rebalanced portfolios. The sample period is from January 2, 1985 to December 29, 1995.

Table 5. Relation between volatility timing and performance

Table 6. Conditional alphas

Interpretation of Table 5. Panel A shows a positive contemporaneous association between volatility timing and the Sharpe ratio. More volatility timers fall into the high Sharpe ratio cell than into the low Sharpe ratio cell, while the opposite is true for non-timers. The evidence is suggestive with $p = 0.0587$. Panel B shows that estimated timing over a six-month period does not predict next-period Sharpe ratios or conditional alphas. Thus, volatility timing is associated with current performance, but the estimated timing coefficient is not a reliable near-term fund selection variable.

Interpretation of Table 6. Conditional alpha is higher when conditional volatility is high. The coefficient ϕ_{mp} is positive for mutual funds in all three model specifications and significantly exceeds the random control sample coefficient. The four-index difference is 0.248 with a strong t -statistic. This result supports the paper's central performance interpretation: funds appear to provide a volatility-state hedge, earning better abnormal performance in high-volatility states when they also lower market exposure.

7.4. Table 7. Volatility and market timing results; Table B.1. Summary of EGARCH(1,1) model of daily S&P 500 returns

Table 7
Volatility and market timing results

Daily		Monthly		
Volatility γ_{mp}	Treyner-Mazuy ϕ_p	Volatility γ_{mp}	Treyner-Mazuy ϕ_p	
			Realized market	Conditional market
−3.23 (−6.00)**		−3.78 (−7.62)**		
−2.91 (−5.80)**	0.20 (3.30)**	−4.25 (−6.12)**	−0.12 (−1.41)	
		−4.49 (−7.84)**		−1.34 (−6.16)

The table shows the daily and monthly volatility coefficients for a single-index specification without a market return term:

$$\text{Daily} \quad R_{pt} = \alpha_p + \beta_{mp0} R_{mt} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \beta_{lmp} R_{m,t-1} + \varepsilon_{pt} \quad (13)$$

$$\text{Monthly} \quad \beta_{mpt} = \beta_{mp0} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) + \varepsilon_{mpt},$$

and with a Treynor–Mazuy (TM) market return term using the actual market returns:

$$\text{Daily} \quad R_{pt} = \alpha_p + \beta_{mp0} R_{mt} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) R_{mt} + \phi_p R_{mt}^2 + \beta_{lmp} R_{m,t-1} + \varepsilon_{pt} \quad (29)$$

$$\text{Monthly} \quad \beta_{mpt} = \beta_{mp0} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) + \phi_p R_{mt} + \varepsilon_{mpt}, \quad (28)$$

and using the conditional market returns

$$\text{Monthly} \quad \beta_{mpt} = \beta_{mp0} + \gamma_{mp} (\sigma_{mt} - \bar{\sigma}_m) + \phi_p R_{mt}^c + \varepsilon_{mpt}. \quad (31)$$

I compute the conditional market return as a function of five known instruments,

$$R_{mt}^c = R_{m0}^c + \sum_{i=1}^k \phi_i z_{i,t-1}. \quad (30)$$

The data represent the difference between the average of all mutual funds and a random control sample. I generate the daily conditional volatility estimates from an EGARCH(1,1) with leverage specification of S&P 500 returns. I take the monthly volatility estimate for month t as the standard deviation of the daily returns that fall within month t . t -statistics are in parenthesis. I base

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Table B.1
Summary of EGARCH(1,1) model of daily S&P 500 returns

Panel A: Estimated coefficients

Parameter	Value	t -statistic
c	0.00048	3.79
a	−0.436	−5.23
a_1	0.093	5.76
b_1	0.966	133
γ_1	−0.538	−3.73

Panel B: Test statistics

	Jarque-Bera normality test for S&P 500 returns	Ljung-Box test for standardized residuals	Ljung-Box test for squared standardized residuals
Statistic	6692	17.08	8.16
p -value	0.00	0.15	0.77

Panel A shows the estimated coefficients and test statistics for an EGARCH(1,1) with leverage term specification of daily S&P 500 returns:

$$\begin{aligned} r_{mt} - r_{ft} &= c_m + \varepsilon_{mt} \\ \varepsilon_{mt} | \varepsilon_{m,t-1}, \varepsilon_{m,t-2}, \dots &\sim t(0, \sigma_{mt}^2) \\ \ln \sigma_{mt}^2 &= a + a_1 \frac{|\varepsilon_{m,t-1}| + \gamma_1 \varepsilon_{m,t-1}}{\sigma_{m,t-1}} + b_1 \ln \sigma_{m,t-1}^2 \end{aligned} \quad (19)$$

Panel B shows the test statistics associated with the EGARCH estimation. I assume error terms are t -distributed. The sample period is from January 2, 1985 through December 29, 1995.

Table 7. Volatility and market timing results

Table B.1. Summary of EGARCH(1,1) model of daily S&P 500 returns

Interpretation of Table 7. Adding Treynor-Mazuy return timing terms does not remove the volatility timing effect. The daily volatility coefficient remains significantly negative after including a squared market return term. The monthly volatility coefficient also remains significant when using realized market returns or conditional market returns based on public instruments. This shows that the volatility timing result is not simply return timing in disguise.

Interpretation of Table B.1. The EGARCH model displays strong volatility persistence through $b_1 = 0.966$ and a leverage/asymmetry effect through $\gamma_1 = -0.538$. The residual diagnostic statistics suggest that the EGARCH specification captures much of the serial dependence in residuals and squared residuals, though the Jarque-Bera statistic rejects normality. This matters because the conditional volatility series is a key input to the daily volatility timing regressions.

8. What the tables jointly establish

The tables build a consistent empirical chain. Table 1 defines the sample and shows meaningful differences between survivors and nonsurvivors. Tables 2 and 3 establish the main daily and monthly volatility timing result. Table 4 clarifies that the evidence does not require private volatility forecasting. Tables 5 and 6 link timing to performance, with a stronger state-contingent alpha result than predictive fund-selection result. Table 7 rules out a simple return-timing explanation. Table B.1 documents the conditional volatility model underlying the daily tests.

9. Limitations and cautions

- The evidence is strongest for the existence of beta-volatility conditioning, not for private volatility forecasting skill.
- Daily mutual fund returns can be affected by net asset value smoothing, stale prices, and nonsynchronous trading. Busse mitigates this with lagged factor returns and robustness checks, but these issues remain relevant.
- The paper infers timing from return regressions rather than directly observing daily portfolio holdings, derivatives positions, or cash allocations.
- The survivor/non-survivor contrast is suggestive but not a causal proof that volatility timing causes survival.
- Conditional alphas depend on the chosen factor model. The paper checks several models, but model dependence is inherent in risk-adjusted performance measurement.

10. Concise summary

Busse shows that actively managed equity mutual funds lower market beta when market volatility is high. This pattern is not a passive artifact, since it is stronger than in style-matched random portfolios. It is strongest for surviving and growth-oriented funds. The evidence does not show strong monthly private volatility forecasting beyond public models, but it does show that volatility timing is associated with better contemporaneous risk-adjusted performance and higher conditional alphas in high-volatility states.